Parallel & Distributed Computing

Lab 2

Single-Node Docker Bridge Networking

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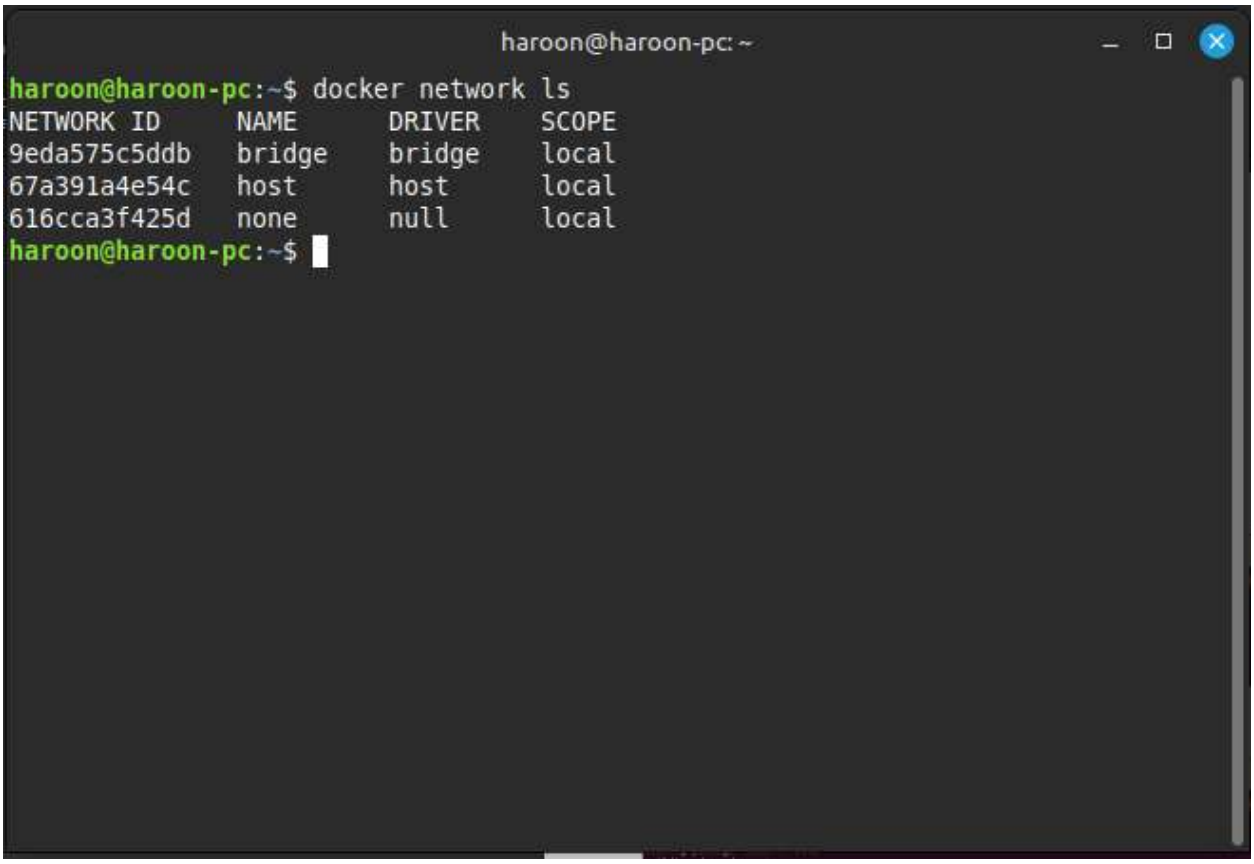
Verify available networks on the docker host

docker network ls



Install the brctl utility to use the Linux bridges on Docker host

sudo apt-get install bridge-utils

```
haroon@haroon-pc: ~
9eda575c5ddb   bridge    bridge    local
67a391a4e54c   host      host      local
616cca3f425d   none      null      local
haroon@haroon-pc:~$ sudo apt-get install bridge-utils
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  ifupdown
The following NEW packages will be installed:
  bridge-utils
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 33.9 kB of archives.
After this operation, 118 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 bridge-utils amd64 1.7.1
-1ubuntu2 [33.9 kB]
Fetched 33.9 kB in 1s (27.2 kB/s)
Selecting previously unselected package bridge-utils.
(Reading database ... 574889 files and directories currently installed.)
Preparing to unpack .../bridge-utils_1.7.1-1ubuntu2_amd64.deb ...
Unpacking bridge-utils (1.7.1-1ubuntu2) ...
Setting up bridge-utils (1.7.1-1ubuntu2) ...
Processing triggers for man-db (2.12.0-4build2) ...
haroon@haroon-pc:~$
```

Create a Bridge Network called **bda-net**

sudo docker network create −−driver bridge bda−net

```
haroon@haroon-pc: ~
 Insecure Registries:
  hubproxy.docker.internal:5555
  ::1/128
  127.0.0.0/8
 Live Restore Enabled: false

haroon@haroon-pc:~$ docker network create --driver bridge bda-net
2d2286ff5683cf252c8900b8954f1e4a53f3094b4a61217b305a3b4ff13e07b9
haroon@haroon-pc:~$ brctl show
haroon@haroon-pc:~$ docker run -dit --name ubuntu-1 ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
817807f3c64e: Pull complete
e4a1e8de092c: Download complete
Digest: sha256:186072bba1b2f436cbb91ef2567abca677337cfc786c86e107d25b7072feef0c
Status: Downloaded newer image for ubuntu:latest
40ae13c2a50077382ec8c66baf984bb01bba5500a4e8358f6698675319cfb992
haroon@haroon-pc:~$ docker run -dit --name ubuntu-2 ubuntu
d1832681927ce2d868fb18fd05115b8046f159b2bf31fb27bcbe4753a1f885a1
haroon@haroon-pc:~$
```

Create 2 Containers ubuntu-1 and ubuntu-2

sudo docker run –dit ––name ubuntu-1 ubuntu

sudo docker run –dit ––name ubuntu-2 ubuntu


```
haroon@haroon-pc: ~
 Insecure Registries:
  hubproxy.docker.internal:5555
  ::1/128
  127.0.0.0/8
 Live Restore Enabled: false

haroon@haroon-pc:~$ docker network create --driver bridge bda-net
2d2286ff5683cf252c8900b8954f1e4a53f3094b4a61217b305a3b4ff13e07b9
haroon@haroon-pc:~$ brctl show
haroon@haroon-pc:~$ docker run -dit --name ubuntu-1 ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
817807f3c64e: Pull complete
e4a1e8de092c: Download complete
Digest: sha256:186072bba1b2f436cbb91ef2567abca677337cfc786c86e107d25b7072feef0c
Status: Downloaded newer image for ubuntu:latest
40ae13c2a50077382ec8c66baf984bb01bba5500a4e8358f6698675319cfb992
haroon@haroon-pc:~$ docker run -dit --name ubuntu-2 ubuntu
d1832681927ce2d868fb18fd05115b8046f159b2bf31fb27bcbe4753a1f885a1
haroon@haroon-pc:~$
```


Verify Containers

docker container ls

```
haroon@haroon-pc:~$ docker container ls
CONTAINER ID   IMAGE     COMMAND       CREATED          STATUS          PORTS     NAMES
d1832681927c   ubuntu    "/bin/bash"   12 minutes ago   Up 12 minutes             ubuntu-2
40ae13c2a500   ubuntu    "/bin/bash"   12 minutes ago   Up 12 minutes             ubuntu-1
```

Connect both containers (**ubuntu-1** and **ubuntu-2)** to bridge network (**bda-net**)

docker network connect bda-net ubuntu-1

docker network connect bda-net ubuntu-2

```
haroon@haroon-pc:~$ docker network connect bda-net ubuntu-1
haroon@haroon-pc:~$ docker network connect bda-net ubuntu-2
haroon@haroon-pc:~$
```

Verify both containers are connected with **bda-net**

docker network inspect bda-net

```
haroon@haroon-pc:~$ docker network inspect bda-net
[
    {
        "Name": "bda-net",
        "Id": "2d2286ff5683cf252c8900b8954f1e4a53f3094b4a61217b305a3b4ff13e07b9",
        "Created": "2026-03-26T08:34:38.268612769Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.18.0.0/16",
                    "Gateway": "172.18.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {},
        "Containers": {
            "40ae13c2a50077382ec8c66baf984bb01bba5500a4e8358f6698675319cfb992": {
                "Name": "ubuntu-1",
                "EndpointID": "c000f74347e34786c10efd3f0ab04ca01eea45955f993ddd5608ad09bcbbd53d",
                "MacAddress": "e6:d5:8e:06:67:47",
                "IPv4Address": "172.18.0.2/16",
                "IPv6Address": ""
```

```
haroon@haroon-pc: ~
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {},
        "Containers": {
            "40ae13c2a50077382ec8c66baf984bb01bba5500a4e8358f6698675319cfb992": {
                "Name": "ubuntu-1",
                "EndpointID": "c000f74347e34786c10efd3f0ab04ca01eea45955f993ddd5608ad09bcbbd53d",
                "MacAddress": "e6:d5:8e:06:67:47",
                "IPv4Address": "172.18.0.2/16",
                "IPv6Address": ""
            },
            "d1832681927ce2d868fb18fd05115b8046f159b2bf31fb27bcbe4753a1f885a1": {
                "Name": "ubuntu-2",
                "EndpointID": "155efd43de9b78e576e9f37390472e4caebcb42313a217f5239d0576bef33cf1",
                "MacAddress": "e6:3d:bb:a3:14:08",
                "IPv4Address": "172.18.0.3/16",
                "IPv6Address": ""
            }
        },
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.18.0.0/16": {
                        "IPsInUse": 5,
                        "DynamicIPsAvailable": 65531
                    }
                }
            }
        }
    }
]
haroon@haroon-pc:~$
```

Exec into the container and Install the **iputils-ping** package in the container

sudo docker exec -it ubuntu-1 bash

apt-get update && apt-get install iputils-ping -y

```
                                                                    root@40ae13c2a500:/
haroon@haroon-pc:~$ docker exec -it ubuntu-1 bash
root@40ae13c2a500:/# apt update && apt install iputils-ping -y
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1954 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [3429 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [34.8 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1504 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2334 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [2154 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [38.1 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3601 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [49.5 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [36.1 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [695 B]
Fetched 37.3 MB in 17s (2181 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
10 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcap2-bin libpam-cap
The following NEW packages will be installed:
  iputils-ping libcap2-bin libpam-cap
0 upgraded, 3 newly installed, 0 to remove and 10 not upgraded.
Need to get 91.2 kB of archives.
After this operation, 322 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcap2-bin amd64 1:2.66-5ubuntu2.2 [34.2 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 iputils-ping amd64 3:20240117-1ubuntu0.1 [44.6 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libpam-cap amd64 1:2.66-5ubuntu2.2 [12.5 kB]
Fetched 91.2 kB in 2s (50.2 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
```

```
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libcap2-bin.
(Reading database ... 4381 files and directories currently installed.)
Preparing to unpack .../libcap2-bin_1%3a2.66-5ubuntu2.2_amd64.deb ...
Unpacking libcap2-bin (1:2.66-5ubuntu2.2) ...
Selecting previously unselected package iputils-ping.
Preparing to unpack .../iputils-ping_3%3a20240117-1ubuntu0.1_amd64.deb ...
Unpacking iputils-ping (3:20240117-1ubuntu0.1) ...
Selecting previously unselected package libpam-cap:amd64.
Preparing to unpack .../libpam-cap_1%3a2.66-5ubuntu2.2_amd64.deb ...
Unpacking libpam-cap:amd64 (1:2.66-5ubuntu2.2) ...
Setting up libcap2-bin (1:2.66-5ubuntu2.2) ...
Setting up libpam-cap:amd64 (1:2.66-5ubuntu2.2) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl
/5.38.2 /usr/local/share/perl/5.38.2 /usr/lib/x86_64-linux-gnu/perl5/5.38 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.38 /usr/s
hare/perl/5.38 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 8.)
debconf: falling back to frontend: Teletype
```

Verify communication between containers (**ubuntu-1 and ubuntu-2**)

ping -c 3 172.18.0.3

```
root@40ae13c2a500:/# ping -c 3 172.18.0.3
PING 172.18.0.3 (172.18.0.3) 56(84) bytes of data.
64 bytes from 172.18.0.3: icmp_seq=1 ttl=64 time=0.298 ms
64 bytes from 172.18.0.3: icmp_seq=2 ttl=64 time=0.477 ms
64 bytes from 172.18.0.3: icmp_seq=3 ttl=64 time=0.197 ms

--- 172.18.0.3 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2032ms
rtt min/avg/max/mdev = 0.197/0.324/0.477/0.115 ms
root@40ae13c2a500:/#
```